

MedFlow Caribbean

Phase 4 — Clinical algorithm layer, with biliary disease worked end-to-end

This document plans Phase 4 of MedFlow Caribbean: a pathway-driven clinical reasoning layer that guides information gathering and management decisions, deliberately pruning superfluous actions that do not change patient outcome. It defines the architecture of the algorithm engine, the data model every pathway uses, the integration with the scoring layer and sign-off pattern from Phase 3, and works the suspected-biliary-disease pathway end-to-end as the canonical reference implementation.

The clinical premise

Most EMRs collect every conceivable field at every encounter. The result: long documentation times, clinician burnout, and noise that obscures the data that actually changes management. Phase 4 takes the opposite stance. At each step, MedFlow asks only what is needed to advance the differential or change management, and explicitly justifies each ask in terms of a downstream decision. If a question or test cannot name the management decision it influences, it does not belong in the pathway. This is the Choosing Wisely framing made executable: evidence-anchored, outcome-driven, low-noise documentation.

The architectural premise

Phase 4 sits between Layer A (deterministic scoring) and Layer B (AI augmentation) from the Phase 3 plan. The core of every pathway is deterministic — a tree of decision nodes encoded as JSON, no AI required, no PHI leaves the device. AI is invoked optionally at specific marked nodes where clinical judgement under uncertainty would normally apply (atypical presentations, conflicting findings, complex comorbidities) and always renders through the same suggestion-chip pattern with full audit logging.

The contract of the algorithm layer

Every node in every pathway carries an evidence anchor: *why this question, why this test, what management decision it changes*. A clinician can always deviate, override, or skip a node — the algorithm proposes, the clinician disposes. The audit log records what was suggested and what was done. The algorithm never autonomously orders investigations, prescribes medication, or books procedures; it presents the next step as a chip the clinician explicitly approves.

Section 1 — Algorithm layer architecture

Three kinds of algorithm content sit inside the layer. Each has a different shape and is rendered slightly differently, but all share the same node vocabulary, the same chip-driven UX, and the same audit trail.

1. Clinical decision rules

Validated rules with binary action outputs. Wells score for PE (CT-PA or not), NEXUS for cervical spine (image or not), Ottawa rules for ankle and knee injury, San Francisco syncope rule (admit or discharge), HEART for chest pain (admit or discharge), ABCD2 for TIA, Centor for streptococcal pharyngitis, Alvarado for appendicitis. Architecturally these are scoring cards (Phase 3 Section 1) with an action chip on the output rather than a severity band. Implementation: extend the existing ScoreCard component with an optional actionOutputs field.

2. Diagnostic pathways

Structured workups for common presentations rendered as decision trees. Suspected biliary disease (this document's worked example). Suspected acute abdomen. Suspected acute coronary syndrome. Suspected stroke. Each pathway has an entry node defined by triggering symptoms, a sequence of question and investigation nodes, branch points where the differential narrows, and outcome nodes that hand off to a management algorithm. The hard work is encoding pathways that reflect current evidence and the local care context (Saint Lucia ERCP availability at Tapion, MRCP access, ICU bed availability, etc).

3. Management algorithms

What to do once diagnosis is made. TG18-graded cholangitis → ERCP timing. Acute pancreatitis fluid resuscitation. DKA management. Acute coronary syndrome thrombolysis vs PCI. These are decision trees with treatment outputs. Same node vocabulary; outputs are orders, prescriptions, procedure bookings, or referrals — always rendered as chips for explicit clinician approval.

Node taxonomy

Six node types, plus one optional AI node, cover the full vocabulary:

Type	Purpose	UI rendering
TRIGGER	Entry point defined by presenting features	Chip listing matched symptoms; clinician confirms entry
QUESTION	Specific data collection (history, exam)	Inline form field; auto-pull from intake if available
ACTION	Order an investigation or intervention	Chip with the action; clinician accepts to commit to order
DECISION	Branch based on data already collected	Visualised as a fork; advances automatically once conditions met
SCORE	Invoke a calculator from the scoring layer	Embedded score card; result drives next node
OUTCOME	Terminal node; pathway resolution	Summary chip; recorded in encounter; pathway closed
AI ASSIST	Optional AI-augmented reasoning at ambiguous decision chips	Suggestion chip with explanation; full audit; opt-out per encounter

Section 2 — Pathway data model

Every pathway is a JSON document with a header (id, name, version, evidence citations, scope), an entry node id, and a nodes dictionary. This makes pathways version-controlled, peer-reviewable, exportable as PDFs for clinical team review, and updateable without changing application code. A pathway is a clinical content artifact, not a feature.

Top-level shape

```
{
  "id": "pathway-biliary-disease",
  "name": "Suspected biliary disease",
  "version": "1.0.0",
  "author": "DDK",
  "reviewedAt": "2026-05-10",
  "evidence": [
    "Tokyo Guidelines 2018 (cholangitis, cholecystitis)",
    "ASGE 2019 suspected choledocholithiasis",
    "WSES 2020 acute cholecystitis guidelines",
    "Revised Atlanta Classification 2012 (pancreatitis)"
  ],
  "scope": "Adult patients with RUQ/epigastric pain, jaundice,\n          fever, or biochemical eviden",
  "entry": "node-B0",
  "nodes": { ... }
}
```

Node shapes

Each node has a common header (id, type, label, evidence anchor, next) plus type-specific fields:

```
// Common header (all nodes)
{
  "id": "node-B1",
  "type": "question" | "action" | "decision" |
    "score" | "outcome" | "trigger" | "ai",
  "label": "Charcot's triad assessment",
  "evidence": "TG18 cholangitis diagnostic criteria",
  "rationale": "Identifies cases requiring biliary drainage",
  "next": "node-B2"
}

// Question node adds:
{ ..., "fields": [
  { "key": "jaundice", "label": "Jaundice?",
    "type": "boolean" },
  { "key": "feverMax", "label": "Max temp (°C)",
    "type": "number", "min": 35, "max": 42 }
] }

// Action node adds:
{ ..., "orders": [
  { "code": "LFT", "label": "Liver function tests" },
  { "code": "USS-ABDO", "label": "Ultrasound abdomen" }
] }

// Decision node replaces 'next' with branches:
{ ..., "branches": [
  { "if": "jaundice && feverMax >= 38",
    "goto": "node-cholangitis-1" },
  { "if": "lipase > 3 * lipaseULN",
    "goto": "node-pancreatitis-1" }
] }

// Score node adds:
{ ..., "calculator": "tokyoCholangitis",
  "resultBindings": { "grade": "tg18Grade" } }

// AI node adds:
{ ..., "prompt": "Given the findings, suggest...",
  "requiresApproval": true,
  "compliance": "requires-baa" }
```

Conditions use a small, safe expression language — not arbitrary JavaScript. Allowed operators: comparison (`==`, `!=`, `<`, `>`, `<=`, `>=`), logic (`&&`, `||`, `!`), references to data collected earlier in the same encounter, and named constants (e.g. `lipaseULN`). This keeps pathways auditable and prevents content authors from accidentally introducing arbitrary code execution.

Section 3 — Engine and UI architecture

Engine

A pure function `advancePathway(state, action)` consumes the current pathway state and a clinician action (accept / edit / reject / override / skip) and returns the new state. Pathway state contains: the active pathway id, the current node id, a history of visited nodes with their resolutions, the data collected so far (a key-value bag), and a reference to the audit log entries generated during this pathway run.

```
// engine/pathway.js
export function advancePathway(state, action) {
  const pathway = getPathway(state.pathwayId);
  const currentNode = pathway.nodes[state.currentNodeId];

  switch (action.type) {
    case "accept":
      return applyResolution(state, currentNode,
        action.payload, "accepted");
    case "edit":
      return applyResolution(state, currentNode,
        action.payload, "edited");
    case "reject":
      return skipNode(state, currentNode, action.reason);
    case "override":
      return jumpToNode(state, action.targetNodeId,
        action.reason);
  }
}

function nextNode(pathway, currentNode, data) {
  if (currentNode.type === "decision") {
    for (const branch of currentNode.branches) {
      if (evaluateCondition(branch.if, data)) {
        return branch.goto;
      }
    }
    return currentNode.defaultGoto || null;
  }
  return currentNode.next;
}
```

UI

The active pathway renders as a *Pathway Drawer* on the right side of the encounter editor, parallel to the Scoring Drawer from Phase 3 and switchable with a tab. The current node shows at the top with its label, rationale, and evidence anchor; the suggested action(s) render as suggestion chips below; visited nodes appear above as a collapsed breadcrumb showing the trail taken. The clinician can scroll up the breadcrumb to revisit and revise any past node — doing so creates a new audit entry but does not destroy the prior decision; both versions remain in the audit trail.

Key UI affordances: **Skip this step** (justification optional but logged); **Why are you asking this?** (opens the rationale and evidence anchor); **Override** (jump to a different node, with a logged reason); **Close pathway** (mark this pathway as not applicable for this encounter, with a reason logged). At no point does the pathway block the clinician from free-text documentation in the standard encounter blocks; the pathway augments, it does not constrain.

Authoring vs runtime separation

Pathway JSON is content, not code. Authoring happens in a separate (eventual) Pathway Studio tool with version control, peer review, and clinical sign-off by a domain expert. Runtime is just the engine + UI. This separation matters because clinical pathways will be edited by clinicians — not just by developers — and the system needs to survive that.

Section 4 — Biliary disease pathway, worked end-to-end

The reference pathway. Covers RUQ pain, jaundice, fever, vomiting, or dark urine in adult patients with suspected biliary origin. Branches into four sub-pathways at the first decision node: acute cholangitis, acute cholecystitis, choledocholithiasis (with or without symptoms), and acute biliary pancreatitis. Evidence anchors: Tokyo Guidelines 2018 (TG18), ASGE 2019 choledocholithiasis criteria, WSES 2020 cholecystitis, Revised Atlanta Classification 2012 for pancreatitis.

Entry triage and initial workup

ID	Type	Label	Content / Action	Next
B0	TRIGGER	Suspected biliary disease	Entry on any of: RUQ or epigastric pain; jaundice; fever $\geq 38^{\circ}\text{C}$ with abdominal symptoms; dark urine; pale stool; recurrent post-prandial pain; biochemical evidence of cholestasis on prior labs. <i>Evidence: Common presenting features; TG18 inclusion criteria</i>	B1
B1	QUESTION	Focused history	Pain location and radiation, duration, character (colicky vs constant), triggers (fatty food). Fever, jaundice, dark urine, pale stool, pruritus. Prior episodes, prior cholecystectomy, prior ERCP. Comorbidities: diabetes, immunosuppression, cirrhosis, biliary stricture. Medications: anticoagulants (affects ERCP timing). Allergies: contrast (affects MRCP and ERCP). <i>Evidence: All items map to subsequent decision nodes; no unused questions</i>	B2
B2	QUESTION	Focused examination	Vitals including temperature. Jaundice (sclera, skin). Abdominal: RUQ tenderness, Murphy's sign, peritonism, palpable gallbladder, hepatomegaly. Mental status (altered = potential Reynolds pentad). Hemodynamic assessment. <i>Evidence: Murphy's drives cholecystitis branch; altered mental status drives Grade III cholangitis branch</i>	B3
B3	ACTION	Mandatory initial investigations	FBC (WBC, neutrophilia). LFTs (bilirubin pattern, ALT/AST/ALP/GGT). Lipase (or amylase if lipase unavailable). U&E; + creatinine. CRP. Coagulation if jaundiced or ERCP anticipated. Blood culture if fever $\geq 38.5^{\circ}\text{C}$. Beta-hCG in women of reproductive age. <i>Evidence: Minimum dataset to populate all branch decisions; no speculative tests</i>	B4
B4	ACTION	First-line imaging	Transabdominal ultrasound, focused biliary protocol: gallbladder wall thickness, stones, pericholecystic fluid, sonographic Murphy's, common bile duct diameter, intrahepatic duct dilation, pancreas (if visualised), free fluid. <i>Evidence: TG18 imaging criterion; sensitivity ~85% for cholelithiasis</i>	B5

B5	DECISION	Branch by clinical pattern	Apply branch conditions in priority order: cholangitis (highest urgency), pancreatitis, cholecystitis, choledocholithiasis without cholangitis, biliary colic. <i>Evidence: Priority order reflects time-to-intervention urgency</i>	see sub-pathways
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Branch conditions at B5 (evaluated top to bottom; first match wins):

- **Cholangitis (C):** Charcot's triad (jaundice + fever + RUQ pain) OR (cholestatic LFTs + fever + WBC elevation + imaging evidence of obstruction). Highest priority.
- **Pancreatitis (P):** Lipase > 3× upper limit of normal, regardless of imaging.
- **Cholecystitis (X):** Murphy's sign + (RUQ tenderness OR fever) + ultrasound features of cholecystitis.
- **Choledocholithiasis (D):** CBD stone on imaging, OR CBD dilation + cholestatic LFTs, without cholangitis or pancreatitis.
- **Biliary colic (L):** Typical symptoms + gallstones on USS + no inflammation, no cholestasis, no other diagnosis.

Sub-pathway C — Acute cholangitis

Highest-urgency branch. Time to biliary drainage is the dominant outcome driver. Mortality in untreated severe cholangitis exceeds 50%; with timely ERCP it falls below 10%.

ID	Type	Label	Content / Action	Next
C1	SCORE	TG18 diagnostic confirmation	Apply TG18 diagnostic criteria: A (systemic inflammation), B (cholestasis), C (imaging). Definite = all three; Suspected = A + (B or C). If suspected only, proceed with treatment but lower threshold for MRCP/imaging review. <i>Evidence: Tokyo Guidelines 2018</i>	C2
C2	SCORE	TG18 severity grading	Invoke Tokyo Guidelines cholangitis severity calculator (Phase 3 Section 1). Inputs: WBC, temperature, age, bilirubin, albumin, organ dysfunction across CV/CNS/Resp/Renal/Hepatic/Hematological systems. Output: Grade I, II, or III. <i>Evidence: TG18 severity criteria; drives drainage timing</i>	C3
C3	DECISION	Severity branch	Grade I → C4 (mild pathway); Grade II → C5 (moderate, early drainage); Grade III → C6 (severe, urgent drainage + ICU). Default if ambiguous: treat as next-higher grade. <i>Evidence: TG18 management algorithm</i>	C4/C5/C6
C4	ACTION	Grade I (mild) — antibiotics, monitor	IV ceftriaxone 2g daily + metronidazole 500mg TDS. Hold po. IV fluids. Reassess in 24h. If no clinical improvement at 24h, escalate to biliary drainage. MRCP if etiology unclear. <i>Evidence: TG18 Grade I management</i>	C7
C5	ACTION	Grade II (moderate) — early drainage	Book ERCP within 24–48h. IV antibiotics as Grade I or escalate to piperacillin-tazobactam if healthcare-associated. Cross-match if endoscopic sphincterotomy anticipated. Coagulation, stop anticoagulants per ESGE bleeding-risk protocol. NPO. Anaesthetic review. <i>Evidence: TG18 Grade II; ESGE periprocedural anticoagulation 2021</i>	C7
C6	ACTION	Grade III (severe) — urgent drainage + ICU	Urgent ERCP within 24h (sooner if available). ICU admission. Organ support per dysfunction: vasopressors, ventilation, RRT. Broad-spectrum antibiotics (piperacillin-tazobactam, or meropenem if local resistance). Blood cultures before antibiotics if not already drawn. Sepsis bundle compliance. <i>Evidence: TG18 Grade III; Surviving Sepsis Campaign 2021</i>	C7

C7	ACTION	ERCP procedural plan	Pre-procedure consent including bleeding, perforation, post-ERCP pancreatitis, infection. Sphincterotomy if stone extraction planned. Stent (plastic vs metal) if drainage required without stone clearance. Rectal indomethacin 100mg pre-procedure for PEP prophylaxis. Document indication, technique, findings, complications in operative note. <i>Evidence: ESGE periprocedural prophylaxis; ASGE quality indicators</i>	C8
C8	OUTCOME	Drainage achieved; continue treatment	Continue antibiotics 4–7 days post-drainage (longer if bacteraemia or incomplete drainage). Repeat LFTs at 48h to confirm trend. Plan definitive treatment: lap cholecystectomy during same admission for gallstone etiology; biliary stricture workup if non-stone cause. <i>Evidence: TG18 follow-up; same-admission cholecystectomy reduces recurrence</i>	(close pathway)

Sub-pathway X — Acute cholecystitis

Most common biliary surgical admission. Outcome driver is timing of cholecystectomy relative to symptom onset.

ID	Type	Label	Content / Action	Next
X1	SCORE	TG18 cholecystitis diagnosis	Apply TG18 cholecystitis diagnostic criteria: local signs (Murphy's, RUQ pain or tenderness or mass), systemic signs (fever, WBC, CRP elevated), imaging (USS features of cholecystitis). Definite ≥ 1 from each. <i>Evidence: Tokyo Guidelines 2018</i>	X2
X2	SCORE	TG18 cholecystitis severity	Invoke TG18 cholecystitis severity calculator (Phase 3 Section 1). Inputs: WBC, duration of symptoms, palpable RUQ mass, marked local inflammation (gangrene, abscess, biliary peritonitis), organ dysfunction. <i>Evidence: TG18 severity criteria</i>	X3
X3	DECISION	Severity and surgical fitness branch	Grade I (mild, fit) → X4 (early lap chole). Grade II (moderate) + fit + expertise available → X4. Grade II + unfit OR expertise unavailable → X5 (cholecystostomy first). Grade III (severe) → X5 (drainage first, delayed surgery). <i>Evidence: TG18 management; WSES 2020</i>	X4/X5
X4	ACTION	Early laparoscopic cholecystectomy	Book lap chole within 72h of symptom onset, ideally same admission. IV antibiotics (ceftriaxone + metronidazole) until source control. ASA assessment. Pre-op imaging review: any suspicion of CBD stone → add MRCP or intraoperative cholangiogram. Consent including conversion, bile duct injury, retained stone. <i>Evidence: WSES 2020 cholecystitis; CHOCOLATE trial</i>	X6
X5	ACTION	Gallbladder drainage; delayed surgery	Percutaneous cholecystostomy (PC) under USS or CT guidance. Or, if PC unavailable, endoscopic transpapillary drainage. IV antibiotics. Optimise comorbidities. Plan delayed interval cholecystectomy 6–8 weeks if patient becomes fit, OR catheter removal study and conservative if remains unfit. <i>Evidence: TG18 management; CHOCOLATE-2 trial</i>	X6
X6	OUTCOME	Definitive treatment completed	If lap chole done: histology of gallbladder, follow-up at 4–6 weeks, return-to-work plan, dietary advice. If cholecystostomy only: plan and date for definitive surgery or catheter management review. <i>Evidence: Standard postoperative pathway</i>	(close pathway)

Sub-pathway D — Choledocholithiasis

CBD stone without cholangitis or pancreatitis. Decision-making centres on probability of stone and timing of intervention.

ID	Type	Label	Content / Action	Next
D1	SCORE	ASGE 2019 probability	Stratify suspected choledocholithiasis probability: High (CBD stone on USS, OR cholangitis, OR bilirubin >4 + dilated CBD) → ERCP. Intermediate (abnormal LFTs OR CBD >6mm OR age >55) → MRCP or EUS. Low (none of the above) → no further workup needed for stone. <i>Evidence: ASGE 2019 suspected choledocholithiasis criteria</i>	D2
D2	DECISION	Probability branch	High → D3 (ERCP direct). Intermediate → D4 (MRCP or EUS first). Low → D5 (no further imaging; proceed to chole if gallstones present). <i>Evidence: ASGE 2019 algorithm</i>	D3/D4/D5
D3	ACTION	High probability — ERCP	Plan ERCP within 24–72h depending on clinical picture. Sphincterotomy, balloon tawl, stent if incomplete clearance. Same procedural considerations as cholangitis pathway C7. If gallstones present and patient fit: plan lap chole same admission (rendezvous procedure or staged). <i>Evidence: ASGE 2019; same-admission chole reduces recurrence</i>	D6
D4	ACTION	Intermediate — MRCP first	MRCP (or EUS if available; EUS has higher sensitivity for small stones). If stone confirmed → jump to D3. If clear → D5. If equivocal, repeat in 48h or proceed to ERCP with diagnostic intent + therapeutic capability. <i>Evidence: ASGE 2019; MRCP sensitivity ~93% for CBD stones</i>	D3 or D5
D5	ACTION	Low/cleared — manage gallbladder	If gallstones present in gallbladder: lap chole on elective list (or same admission if symptomatic). With intraoperative cholangiogram if CBD remains suspicious. If no gallstones and CBD clear: investigate for non-stone cause (stricture, malignancy, parasitic, autoimmune cholangiopathy). <i>Evidence: Standard surgical management</i>	D6
D6	OUTCOME	Pathway resolution	Document stone clearance status, gallbladder management plan, and follow-up imaging if applicable. Recurrence risk counselling. Histology of gallbladder when cholecystectomy performed. <i>Evidence: Standard documentation</i>	(close pathway)

Sub-pathway P — Acute biliary pancreatitis

Severity assessment and early ERCP only when biliary obstruction or cholangitis co-exists, are the two key outcome drivers.

ID	Type	Label	Content / Action	Next
P1	ACTION	Diagnosis confirmed	Revised Atlanta criteria: ≥ 2 of: characteristic abdominal pain; lipase $>3\times$ ULN; imaging features of pancreatitis. Already confirmed at branch entry from B5. <i>Evidence: Revised Atlanta Classification 2012</i>	P2
P2	SCORE	Severity — multiple scores	Calculate BISAP within 24h (Phase 3 score). Calculate Ranson on admission and 48h. Calculate modified Glasgow-Imrie at 48h. Use the most pessimistic to flag for higher care. CT severity index (Balthazar) only if not improving by 72–96h. <i>Evidence: Revised Atlanta + Banks 2013 review; CT not in first 72h unless complications</i>	P3
P3	DECISION	Severity-driven branch	Mild (Atlanta Mild AP, all scores low) → P4. Moderately severe (transient organ failure or local complications) → P5. Severe (persistent organ failure $>48h$) → P6. <i>Evidence: Revised Atlanta classification</i>	P4/P5/P6
P4	ACTION	Mild AP — ward management	Goal-directed IV fluids (lactated Ringer's, 5–10 ml/kg/h, titrated to urine output and lactate). Pain control (paracetamol + opioid). Early enteral nutrition (oral or NJ within 24–48h once tolerating). Anti-emetics. Monitor for worsening. <i>Evidence: WATERFALL trial 2022 (moderate fluid resus); IAP/APA 2013</i>	P7
P5	ACTION	Moderately severe — HDU and supportive	HDU monitoring. Aggressive but goal-directed fluid resuscitation. Early enteral nutrition (NJ preferred if oral not tolerated). Antibiotics only if infected necrosis confirmed (not prophylactic). CT at 72–96h if not improving to assess for necrosis or fluid collections. <i>Evidence: Revised Atlanta; PROPATRIA trial against prophylactic abx</i>	P7
P6	ACTION	Severe AP — ICU and full support	ICU admission. Organ support per organ failure. Goal-directed resuscitation. Enteral nutrition as tolerated, parenteral if needed. Antibiotics only for confirmed or strongly suspected infection. Consider transfer if local ICU capacity exceeded. <i>Evidence: Revised Atlanta severe AP definition</i>	P7
P7	DECISION	Cholangitis co-existence?	Is there evidence of co-existing cholangitis (fever + jaundice + cholestatic LFTs, or persistent CBD obstruction)? <i>Evidence: IAP/APA 2013; ERCP only indicated in biliary AP with cholangitis or persistent obstruction</i>	P8 or P9

P8	ACTION	Yes — urgent ERCP	ERCP within 24–72h depending on cholangitis severity. Hands off to cholangitis sub-pathway C5/C6 for procedural specifics. <i>Evidence: IAP/APA 2013; APEC trial</i>	P9
P9	ACTION	Cholecystectomy plan	For mild biliary AP: same-admission cholecystectomy strongly recommended (reduces recurrence). For moderately severe or severe: delayed cholecystectomy after recovery (typically 4–6 weeks). Document plan; ensure follow-up is booked before discharge. <i>Evidence: PONCHO trial 2015; recurrence ~25% if delayed >2 weeks</i>	P10
P10	OUTCOME	Pathway resolution	Document final Atlanta severity, complications, interventions, and follow-up plan. Etiology summary: gallstones confirmed; consider lipids and other workup if no clear cause despite biliary pathway entry. <i>Evidence: Standard documentation</i>	(close pathway)

Section 5 — Evidence anchoring

Every node in every pathway carries an evidence field. This is non-negotiable. The field can reference a clinical guideline (Tokyo Guidelines 2018, ASGE 2019), a randomised trial (PONCHO, APEC, CHOCOLATE), a consensus document (Revised Atlanta), a systematic review, or local practice standard — but *something* must justify the node's presence. A node with no evidence anchor fails review and cannot be merged into the production pathway.

Authoring requires a separate workflow: pathway content draft → internal review by a second clinician → evidence audit (do citations support the logic?) → dry-run on past cases → release. Versioning is strict; the `version` field at the top of each pathway JSON is incremented on every released change, and old versions remain queryable for audit ("what did the algorithm recommend on 12 March 2026?").

Section 6 — AI augmentation pattern

AI is invited into the algorithm only at marked AI nodes, and only where clinical judgement under uncertainty would normally apply. Three specific use cases drive the current design:

Ambiguous branching. When two or more branch conditions partially match, an AI node can be inserted at the decision point. The AI receives the structured data collected so far (without identifiers) and a curated prompt: "Given these findings, which sub-pathway best fits and why? Cite the criteria you weighed." Output renders as a suggestion chip with rationale; clinician accepts, edits the branch choice, or rejects and routes manually.

Atypical presentation. When the data collected so far fails to match any branch condition, an AI node can suggest the most likely differentials and the next diagnostic step. Same chip pattern; same audit log.

Documentation synthesis. At pathway closure, an AI node can synthesise the encounter summary from the structured pathway data, including which branches were taken, which were skipped and why, and the final disposition. Synthesised text is presented for clinician edit and approval before becoming the final note. Lower clinical risk than the branching use cases but still subject to full sign-off.

What AI does not do in this layer.

AI never autonomously orders investigations, prescribes medications, books procedures, or signs off encounters. It never picks a branch on the clinician's behalf. It never speaks first — it is invoked only at marked nodes or by the clinician's explicit request. Every AI output is a suggestion through the Phase 3 chip pattern with full audit logging, and the clinician's decision stands regardless of the AI's recommendation.

Section 7 — Integration with Phase 3

The Phase 3 plan put scoring, sign-off, and AI compliance as the three workstreams of Phase 3. Phase 4 changes nothing about *what* needs to be built but does shift the *why*:

- The **scoring layer** remains the foundation, but its primary consumer is now the pathway engine rather than standalone clinician use. Tokyo Guidelines cholangitis severity is invoked inside node C2 of the biliary pathway. This makes the score-card UI an embeddable rather than always-standalone component.
- The **sign-off pattern** from Phase 3 Section 2 is now the universal interaction primitive: every algorithm node renders as a chip-driven step. This makes the pattern more central, not less.
- The **AI compliance gates** from Phase 3 Section 3 are unchanged in what they require, but the algorithm layer needs them only for AI-marked nodes. The deterministic core of every pathway can ship without any of the Section 3 gates being closed. This is significant: pathway-driven workflow can be delivered to the clinic before BAA/DPA conversations conclude.

Revised priority order across Phase 3 and Phase 4

	Workstream	Notes	Est.
3a	Storage layer (Item 1)	Blocking everything	2 hrs
3b	Scoring layer foundation + first 6 scores	Tokyo cholangitis, Tokyo cholecystitis, Ranson, Glasgow-Imrie, BISAP, ASA	8 hrs
3c	Sign-off pattern + audit log + chip component	Universal interaction primitive	4 hrs
4a	Pathway engine + JSON loader + condition evaluator	Pure functions, no UI yet	6 hrs
4b	Pathway Drawer UI + breadcrumb + suggestion chips	Renders any pathway JSON	8 hrs
4c	Biliary disease pathway JSON — entry + 1 sub-pathway	Cholangitis sub-pathway first (highest ERCP relevance)	6 hrs
4d	Biliary pathway JSON — remaining sub-pathways	Cholecystitis, choledocholithiasis, pancreatitis	10 hrs
4e	Internal clinical dry-run on 10 past cases	Validate pathway against actual clinic data; tune	Variable
3b+	Remaining scoring cards	Child-Pugh, MELD-Na, Caprini, Alvarado, Charlson, Clavien-Dindo, ECOG, Karnofsky	8 hrs
3d	Compliance work (BAA, DPA, counsel) in parallel	Calendar-time, runs alongside coding	Weeks
4f	First AI node — ambiguous branching at biliary B5	Pilot the AI augmentation pattern	6 hrs

Next	Second pathway (acute abdomen, ACS, or stroke)	Reuse engine + UI; just author new JSON	10–12 hrs/pathway
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Section 8 — How to chunk Phase 4

Phase 4 is the most ambitious phase to date. Realistic time to ship a usable biliary pathway end-to-end (engine + UI + content + internal validation): approximately 50–60 hours of focused build time, plus the underlying Phase 3 work, plus clinical authoring time. This is several months of part-time evening and weekend work, or roughly six weeks at one focused session per day with no clinical interruptions — the latter being unrealistic.

Honest recommendation: engage a developer for the engine and UI build (Phase 4a and 4b), retain authoring of pathway content yourself (Phase 4c, 4d, and future pathways) because that's where clinical accuracy lives, and use the Pathway Studio idea as a longer-term goal that lets you author new pathways without writing JSON by hand.

Honest constraints

- **This requires clinical content authoring, not just code.** The hard work in Phase 4 is the pathway JSON, not the engine. Authoring biliary disease end-to-end well takes several focused sessions and a second-clinician review. Anything less and the pathway reflects one person's interpretation of guidelines, which is exactly the failure mode that gives clinical decision support a bad name.
- **Local context matters.** Pathways must reflect Saint Lucia care realities: Tapion ERCP availability, MRCP access, ICU bed capacity, locally-stocked antibiotics, transfer pathways to Martinique for what isn't available locally. A pathway copied verbatim from a US or UK guideline without local adaptation will recommend interventions that cannot be delivered.
- **Pathway maintenance is ongoing.** Guidelines update. TG18 will become TG23 will become TG28. Atlanta will be revised again. New trials will shift the evidence anchors. Each pathway needs a documented review schedule (12–18 months is typical) and a process for incorporating updates.
- **Clinical liability.** A pathway-driven EMR that proposes specific management steps is closer to a medical device, in regulatory terms, than a documentation tool. The Saint Lucia regulatory framework for clinical decision support software needs counsel's review before this layer is used on real patients. Saint Lucia DPA 2011 review and indemnity insurer notification are now more important than they were in Phase 3.
- **Synthetic data still applies.** The deterministic pathway engine works fine on synthetic data and can be validated extensively before any real-patient exposure. The compliance gates from Phase 3 still gate real-patient deployment.
- **Single-developer ceiling reached, confirmed.** Phase 4 is past the point where solo build is sensible. The minimum sustainable team for clinical deployment of a pathway-driven EMR is one engineer for the platform, one clinician-author for content (you), and access to a second clinician for content review. Anything less and the system is a single point of failure.

The line summary.

Build the engine generic, encode pathways as content, ship one specialty-relevant pathway end-to-end (biliary), validate against past cases, iterate. The first pathway is the expensive one; the second is half the work; the tenth is a weekend's authoring. Done well, MedFlow Caribbean becomes the rarest of clinical tools: one that asks only what changes the decision, justifies every ask, and lets the clinician concentrate on the patient.